OKLAHOMA SCHOOL TESTING PROGRAM

TEST BLUEPRINT AND ITEM SPECIFICATIONS

GRADE 8 MATHEMATICS

OKLAHOMA Education

1 **Which of the following is equivalent to the expression below?**

$$\frac{4^8}{4^2}$$

A 4^4

B 4^6

C 4^{10}

D 4^{16}

Standard: PA.N.1.1 Develop and apply the properties of integer exponents, including $a^0 = 1$ (with $a \neq 0$), to generate equivalent numerical and algebraic expressions.

Depth-of-Knowledge: 1
This item is a DOK 1 because it requires the student to perform a simple one-step procedure, dividing with exponents.

Distractor Rationale:
A. The student divided the exponents.
B. Correct. The student demonstrated an ability to apply the properties of integer exponents.
C. The student added the exponents.
D. The student multiplied the exponents.

2 The weight of a ruby-throated hummingbird was 3,100 milligrams.

How is this weight written in scientific notation?

A 3.1×10^{-3} milligrams

B 3.1×10^{-2} milligrams

C 3.1×10^{2} milligrams

D 3.1×10^{3} milligrams

Standard: PA.N.1.2 Express and compare approximations of very large and very small numbers using scientific notation.

Depth-of-Knowledge: 1
This item is a DOK 1 because it requires the student to perform a simple procedure, writing a given number in scientific notation.

Distractor Rationale:
A. The student confused -3 and 3.
B. The student counted zeroes and confused -2 and 2.
C. The student counted zeroes.
D. Correct. The student demonstrated an ability to express numbers using scientific notation.

3 **A space shuttle travels at 2.6×10^4 feet per second. An hour is 3.6×10^3 seconds. This expression can be used to find the number of feet the space shuttle travels in an hour.**

$$\boxed{(2.6 \times 10^4)(3.6 \times 10^3)}$$

How many feet does the shuttle travel in an hour?

A 6.2×10^1 feet

B 6.2×10^{12} feet

C 9.36×10^7 feet

D 9.36×10^{12} feet

Standard: PA.N.1.3 Multiply and divide numbers expressed in scientific notation and express the answer in scientific notation.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to multiply numbers expressed in scientific notation.

Distractor Rationale:

A. The student computed 2.6 + 3.6 to get the base number and then computed 4 − 1 to get the exponent.

B. The student computed 2.6 + 3.6 to get the base number and then computed 4 × 3 to get the exponent.

C. Correct. The student demonstrated an ability to multiply numbers expressed in scientific notation.

D. The student computed the base number correctly, but then multiplied instead of added to find the exponent.

4 **A smoothie shop makes \$3.00 for every smoothie sold after subtracting the costs of ingredients and packaging. They have additional costs of \$150.00 per day. Which linear model expresses how much the shop makes, *y*, for selling *x* smoothies in a day?**

A $y = 3x$

B $y = 3x - 150$

C $y = 150x - 3$

D $y = 3x + 150$

Standard: PA.A.1.2 Use linear functions to represent and model mathematical situations.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate a verbal description into an equation.

Distractor Rationale:

A. The student ignored the fixed costs.

B. Correct. The student demonstrated an ability to write a linear function to represent a real-world problem.

C. The student reversed the slope and *y*-intercept.

D. The student added the fixed costs instead of subtracting.

5 **A taxi company charges a base fee of \$2.00 plus \$0.50 per mile. Which of these represents the cost of a taxi ride, *y*, for the distance of *x* miles?**

A $y = 2.5x$

B $y = 50x + 2$

C $y = 0.50x + 2$

D $y = 2x + 0.50$

Standard: PA.A.1.2 Use linear functions to represent and model mathematical situations.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate a verbal description into an equation.

Distractor Rationale:

A. The student added the \$2 and the \$0.50.

B. The student confused 50 and 0.50.

C. Correct. The student demonstrated an ability to use a linear function to represent a real-world situation.

D. The student swapped the slope and the *y*-intercept.

6

x	*y*
−7	−12
−5	−8
−3	−4
−1	0
1	4

Which statement best describes the values of *x* and *y* in the table?

A As the value of *x* increases by 2, the value of *y* increases by 4.

B As the value of *x* increases by 2, the value of *y* decreases by 4.

C As the value of *x* increases by 4, the value of *y* increases by 2.

D As the value of *x* increases by 4, the value of *y* decreases by 2.

Standard: PA.A.2.2 Represent linear functions with tables, verbal descriptions, symbols, and graphs; translate from one representation to another.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to analyze the data presented in the table and then describe the relationship between *x* and *y*.

Distractor Rationale:

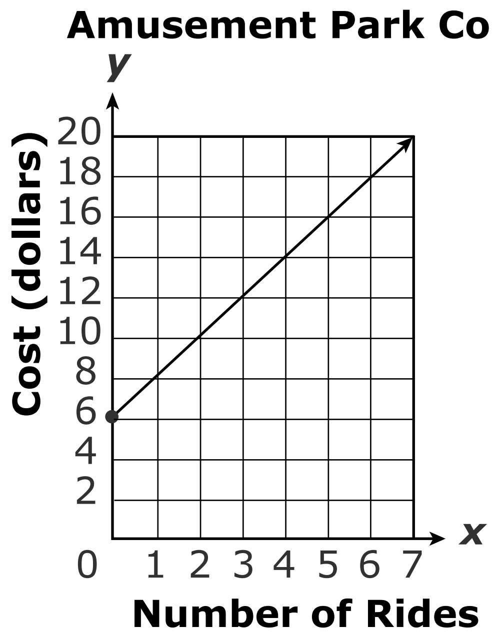
A. **Correct. The student demonstrated an ability to represent a linear function with a verbal description.**

B. The student identified the correct relationship for *x* only.

C. The student reversed *x* and *y*.

D. The student reversed *x* and *y* and thought *y* was decreasing.

7 **When planning a trip to the local amusement park, Leah drew a graph to show her possible costs.**



Based on the graph, what is the cost per ride?

A $1.00

B $2.00

C $2.60

D $6.00

Standard: PA.A.2.3 Identify graphical properties of linear functions including slope and intercepts. Know that the slope equals the rate of change, and that the *y*-intercept is zero when the function represents a proportional relationship.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to analyze the graph to find the cost per ride.

Distractor Rationale:

A. The student saw that the values on the *x*-axis increase by 1 but did not recognize the scale of the *y*-axis is 2.

B. Correct. The student demonstrated an ability to identify graphical properties of a linear function.

C. Balance distractor

D. The student saw that the first data point has a cost of $6.

8

Match the equation in the left column to the description of the slope and *y*-intercept of its graph in the right column. To connect an equation to a description, click an equation in the left column and then a description in the right column, and a line will automatically be drawn between them. To remove a connection, hold the pointer over the line until it turns red, and then click it. Each equation in the left column matches to exactly two descriptions in the right column.

	slope = $-\frac{2}{3}$
	slope = $\frac{2}{3}$
$3x - 2y = 8$	slope = $\frac{3}{2}$
$2x + 3y = 12$	y-intercept = –4
$3y + 6 = 2x$	y-intercept = –2
	y-intercept = 4

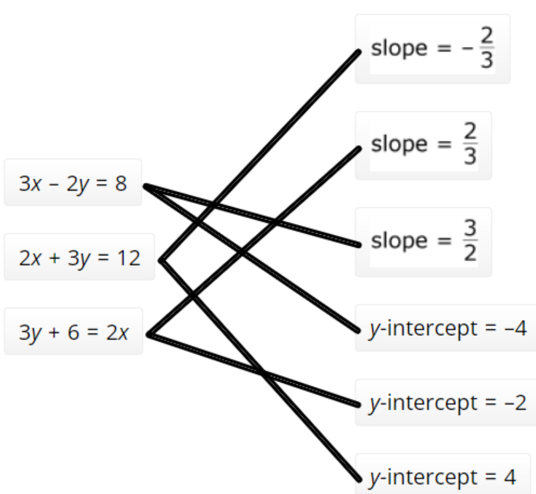
Standard: PA.A.2.3 Identify graphical properties of linear functions including slope and intercepts. Know that the slope equals the rate of change, and that the *y*-intercept is zero when the function represents a proportional relationship.

Depth-of-Knowledge: 1

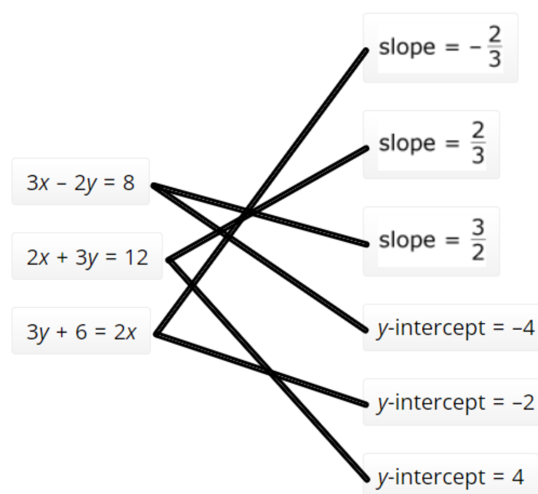
This item is a DOK 1 because it requires the student to identify the slope and *y*-intercept of equations presented in slope-intercept form.

Sample Distractor Rationales:

Correct

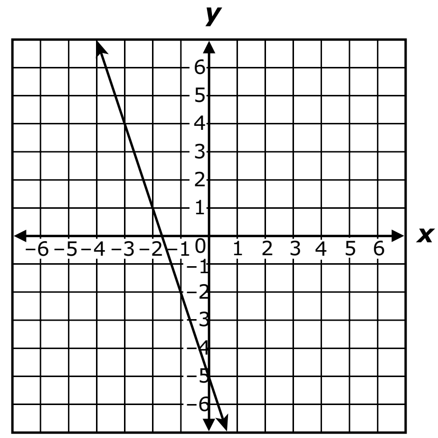


Incorrect



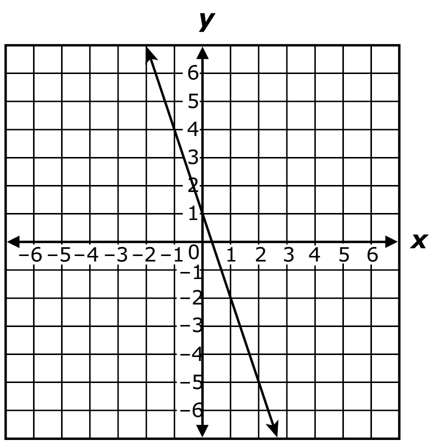
The student confused –2/3 and 2/3.

9 **The graph of the equation $y = -3x - 5$ is shown below.**

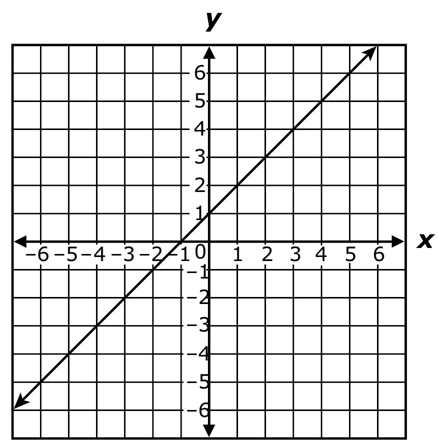


Which <u>best</u> represents the graph of the equation $y = -3x - 5$ when the slope is changed to 1 and the y-intercept remains the same?

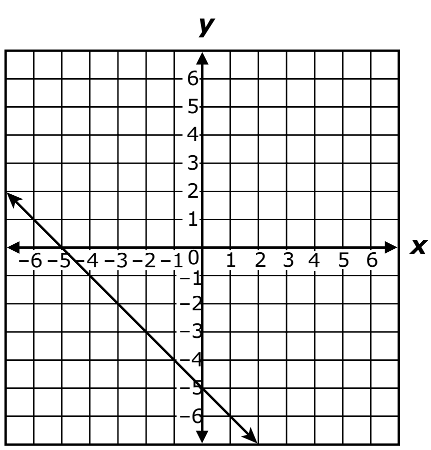
A



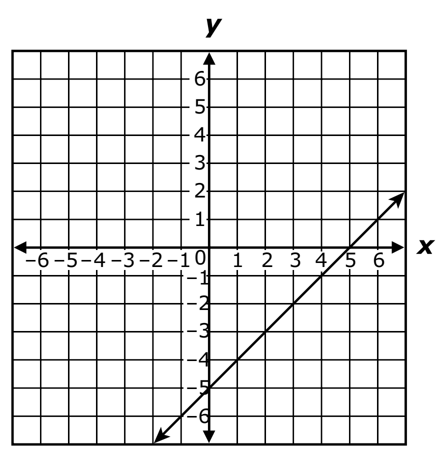
B



C



D



Standard: PA.A.2.4 Predict the effect on the graph of a linear function when the slope or y-intercept changes. Use appropriate tools to examine these effects.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use an understanding of graphs and equations to find a graph that matches an equation with a different slope, but same *y*-intercept.

Distractor Rationale:

A. The student changed the *y*-intercept to 1.
B. The student changed both the slope and the *y*-intercept to 1.
C. The student changed the slope to –1.
D. Correct. The student demonstrated an ability to predict the effect on the graph when the slope changes.

10 A thermometer showed a temperature of 40 degrees Celsius. This equation can be used to find the temperature in degrees Fahrenheit, F, given the temperature in degrees Celsius, C.

$$9C = 5(F - 32)$$

What was the temperature, in degrees Fahrenheit?

A 40°F

B 72°F

C 104°F

D 360°F

Standard: PA.A.2.5 Solve problems involving linear functions and interpret results in the original context.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to solve for a variable in an equation with more than one step.

Distractor Rationale:

A. The student identified the temperature in degrees Celsius.

B. The student computed 40 + 32.

C. Correct. The student demonstrated an ability to solve a problem using a linear function.

D. The student reversed C and F and then multiplied by 9 instead of dividing by 9.

11 **The total amount of money, in dollars, Sandy earns for working *h* hours is represented by this expression.**

$$\boxed{15h}$$

How much money does Sandy earn for working 35 hours?

A $20

B $50

C $525

D $1535

Standard: PA.A.3.1 Use substitution to simplify and evaluate algebraic expressions.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, substituting 35 for *h* in a single step expression.

Distractor Rationale:

A. The student subtracted instead of multiplying.

B. The student added instead of multiplying.

C. Correct. The student demonstrated an ability to use substitution to evaluate an algebraic expression.

D. The student substituted 35 for *h* and then read left to right, not understanding the implied multiplication.

12

Complete these equations. Select the numbers you want to choose and drag and drop them into the boxes. To drag a number, click and hold it, and then drag to the desired box. To change a number, click and hold it, then drag it back to the desired box. You may use each number once, more than once, or not at all.

If $x = 3$, then $5x + 2 = \boxed{}$.

If $x = 1$, then $2(x-4) = \boxed{}$.

If $x = -2$, then $-3x^2 + 4x + 15 = \boxed{}$.

19 17 11 6 −5 −6

Standard: PA.A.3.1 Use substitution to simplify and evaluate algebraic expressions.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, substituting values for *x* expressions.

Sample Distractor Rationales:

Correct

If $x = 3$, then $5x + 2 = 17$.
If $x = 1$, then $2(x - 4) = -6$.
If $x = -2$, then $-3x^2 + 4x + 15 = -5$.

Incorrect

If $x = 3$, then $5x + 2 = 17$.
If $x = 1$, then $2(x - 4)= -6$.
If $x = -2$, then $-3x^2 + 4x + 15 = \mathbf{19}$.

The student thought $(-2)^2$ was -4.

If $x = 3$, then $5x + 2 = 17$.
If $x = 1$, then $2(x - 4) = \mathbf{6}$.
If $x = -2$, then $-3x^2 + 4x + 15 = 19$.

The student computed (1 – 4) as (4 – 1) for the second equation.

13 **The product of 24 and *n* is greater than –96. Which inequality represents the possible values for *n*?**

A $n > -4$

B $n > -120$

C $n < -4$

D $n < -120$

Standard: PA.A.4.2 Represent, write, solve, and graph problems leading to linear inequalities with one variable in the form $px + q > r$ and $px + q < r$, where p, q, and r are rational numbers.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate a verbal description into an inequality when simplifying is necessary.

Distractor Rationale:

A. Correct. The student demonstrated an ability to write a linear inequality with one variable.
B. The student confused sum and product.
C. The student reversed the inequality sign.
D. The student confused sum and product and reversed the inequality sign.

14 Tom has read 11 pages of a 215-page book. He will read 6 pages each day until he finishes the book.

Which equation can be used to find the number of days, ***d***, it will take Tom to finish reading the book?

A $6 + 11d = 215$

B $11 + 6d = 215$

C $17d = 215$

D $6d = 215$

Standard: PA.A.4.3 Represent real-world situations using equations and inequalities involving one variable.

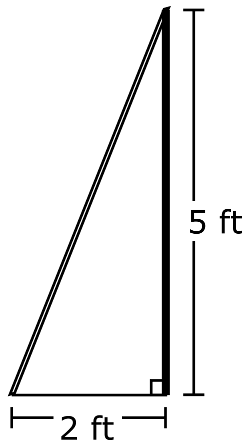
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to represent a real-world situation expressed in words using an equation.

Distractor Rationale:

A. The student confused the slope and the y-intercept.
B. Correct. The student demonstrated an ability to write a linear equation for a real-world problem with one variable.
C. The student did $11d + 6d$.
D. The student ignored the 11 pages already read.

15 **A wire is attached to the top of a 5-foot tall pole. The other end of the wire is secured to the ground 2 feet from the base of the pole.**



What is the length, in feet, of the wire?

A $\sqrt{7}$ feet

B $\sqrt{14}$ feet

C $\sqrt{21}$ feet

D $\sqrt{29}$ feet

Standard: PA.GM.1.1 Justify the Pythagorean theorem using measurements, diagrams, or dynamic software to solve problems in two dimensions involving right triangles.

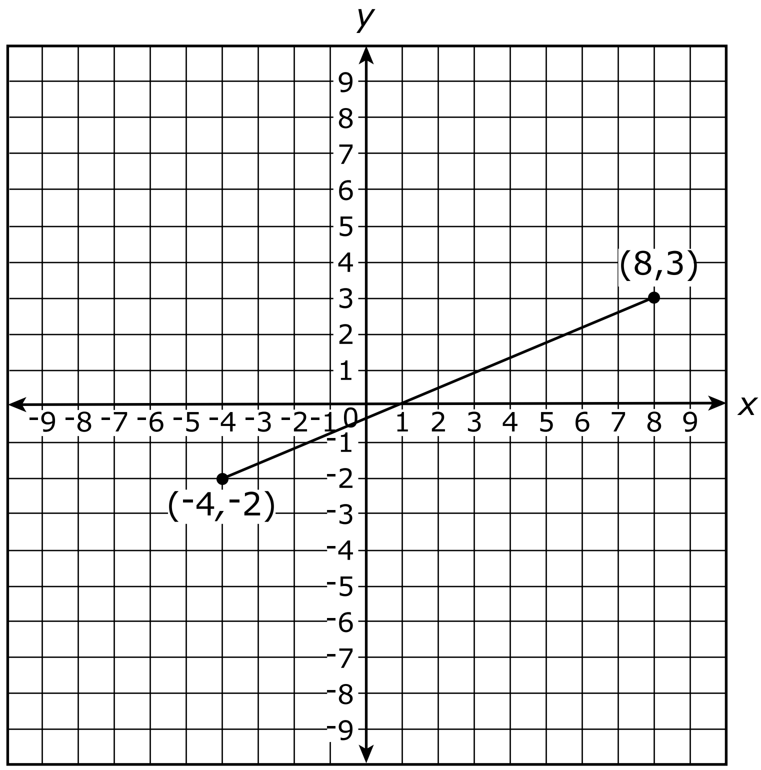
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, using the Pythagorean Theorem to solve for the hypotenuse of a triangle.

Distractor Rationale:

A. The student thought $a + b = c^2$.
B. The student thought $c^2 = 2a + 2b$.
C. Balance distractor
D. Correct. The student demonstrated an ability to use the Pythagorean Theorem to solve a problem involving a right triangle.

16



What is the length of the segment connecting the points that have the coordinates (–4, –2) and (8, 3)?

A 12

B 13

C 14

D 15

Standard: PA.GM.1.2 Use the Pythagorean Theorem to find the distance between any two points in a coordinate plane.

Depth-of-Knowledge: 2

This item is a DOK 2 because the student must first realize that the Pythagorean Theorem is necessary and then utilize it to find the distance between two points on the coordinate plane.

Distractor Rationale:

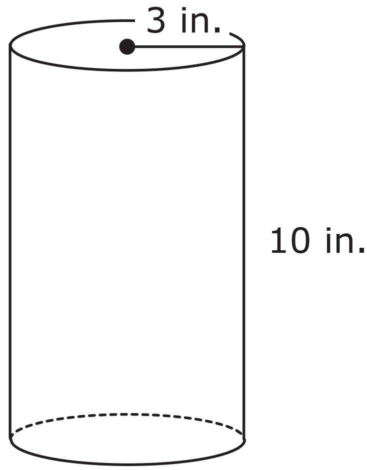
A. The student found the length of the horizontal leg.

B. Correct. The student demonstrated an ability to use the Pythagorean Theorem to the find the distance between two points on the coordinate plane.

C. The student found the difference of the two legs and then multiplied by 2.

D. The student added the lengths of the two legs.

17 **Jason decorated the outside of this cylinder.**



What is the surface area of the cylinder, in square inches?

A 72π square inches

B 78π square inches

C 192π square inches

D 260π square inches

Standard: PA.GM.2.2 Calculate the surface area of a cylinder, in terms of π and using approximations for π, using decomposition or nets. Use appropriate units (e.g., cm^2).

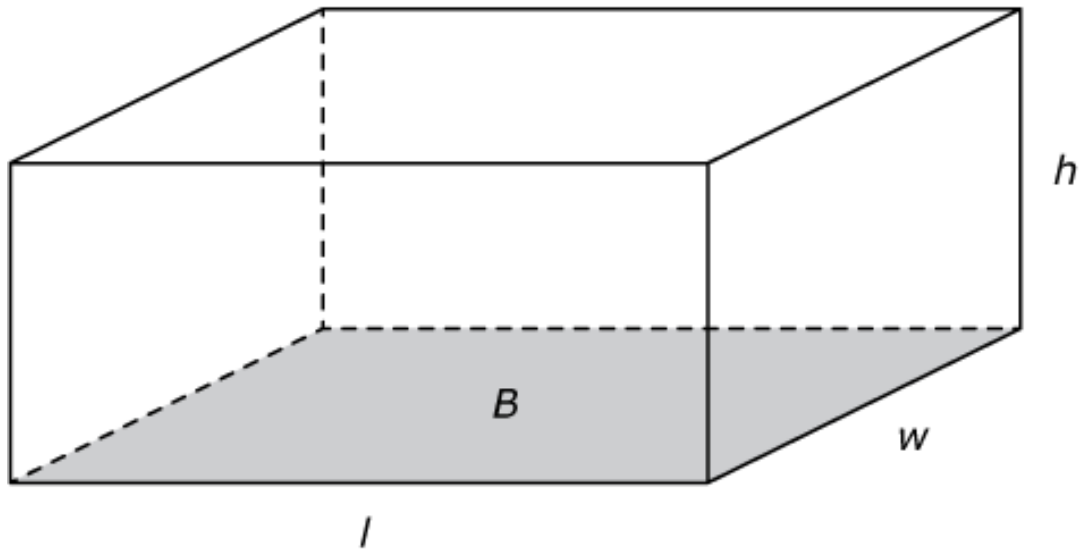
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the surface area of a cylinder.

Distractor Rationale:

A. The student computed 3^2 as 6 instead of 9.
B. Correct. The student demonstrated an ability to calculate the surface area of a cylinder in terms of π.
C. The student used the diameter, 6, instead of the radius, 3.
D. Balance distractor

18 **A right rectangular prism has a base area of 24 cm^2. Its volume, in cubic centimeters, is <u>not</u> a whole number.**



Select the measures that could be two of the dimensions of this prism.
To select a measure, click the measure. To deselect a measure, click it again.

length = 6 cm width = 4 cm	length = 6.1 cm height = 2 cm	width = 3.9 cm height = 8.4 cm
length = 7.5 cm width = 3.5 cm	length = 12 cm height = 15.1 cm	width = 10.5 cm height = 9 cm

Standard: PA.GM.2.3 Justify why base area (*B*) and height (*h*) in the formula $V = Bh$ are multiplied to find the volume of a rectangular prism. Use appropriate units (e.g., cm^3).

Depth-of-Knowledge: 3

This item is a DOK 3 because it requires the student to use the formula for the volume of a right rectangular prism to determine possible measures for the length and width, when the exact volume is not given, but only that it is not a whole number.

Sample Distractor Rationales:

Correct

length = 6 cm width = 4 cm		width = 3.9 cm height = 8.4 cm
	length = 12 cm height = 15.1 cm	

Incorrect

length = 6 cm width = 4 cm		

The student identified only the dimensions that show an area of 24 cm^2.

	length = 6.1 cm height = 2 cm	width = 3.9 cm height = 8.4 cm
length = 7.5 cm width = 3.5 cm	length = 12 cm height = 15.1 cm	width = 10.5 cm height = 9 cm

The student selected all dimensions that included at least one non-whole number.

19 **A cylindrical container has a diameter of 6 centimeters and a height that is $\frac{5}{3}$ times the diameter of the cylinder. What is the volume of the container, in cubic centimeters?**

A 60π cubic centimeters

B 90π cubic centimeters

C 120π cubic centimeters

D 360π cubic centimeters

Standard: PA.GM.2.4 Develop and use the formulas $V = (\pi r)^2 h$ and $V = Bh$ to determine the volume of right cylinders, in terms of π and using approximations for pi (π). Justify why base area (B) and height (h) are multiplied to find the volume of a right cylinder. Use appropriate units (e.g., cm^3).

Depth-of-Knowledge: 3

This item is a DOK 3 because it requires the student to use the formula for the volume of a cylinder when the height is given as a fraction of the diameter.

Distractor Rationale:

A. The student thought 3^2 is 6.

B. Correct. The student demonstrated an ability to use the formula $V = \pi r^2 h$ to determine the volume of a right cylinder.

C. The student used 6 for the radius and thought 6^2 is 12.

D. The student used 6 for the radius.

20 **These data show the ages of students in a community play.**

12, 9, 11, 9, 15, 11, 9, 11, 12

Which data point could be removed from the set without changing the mean or the median of the data?

A 9

B 11

C 12

D 15

Standard: PA.D.1.1 Describe the impact that inserting or deleting a data point has on the mean and the median of a data set. Create data displays using technology to examine this impact.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to have a deep understanding of mean and median in order to determine which data point could be removed from a set of data without changing the mean or median.

Distractor Rationale:

A. The student saw that this will not change the median.

B. Correct. The student demonstrated an ability to describe the impact that deleting a data point has on the mean and the median of a data set.

C. The student saw that this will not change the median.

D. The student saw that this will not change the median.

21 **Brandon used an indoor rock-climbing wall seven times. His climbing times, in minutes, are shown in this list.**

35, 16, 17, 18, 13, 13, 14

Why is the median the most useful measure of central tendency for these times?

A The median is not affected by an outlier.

B The median is equal to the range of the data.

C The median is the time that occurs most often.

D The median is a larger value than the mean of the data.

Standard: PA.D.1.2 Explain how outliers affect measures of center and spread.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to identify why the median is the best measure for a given set of data.

Distractor Rationale:

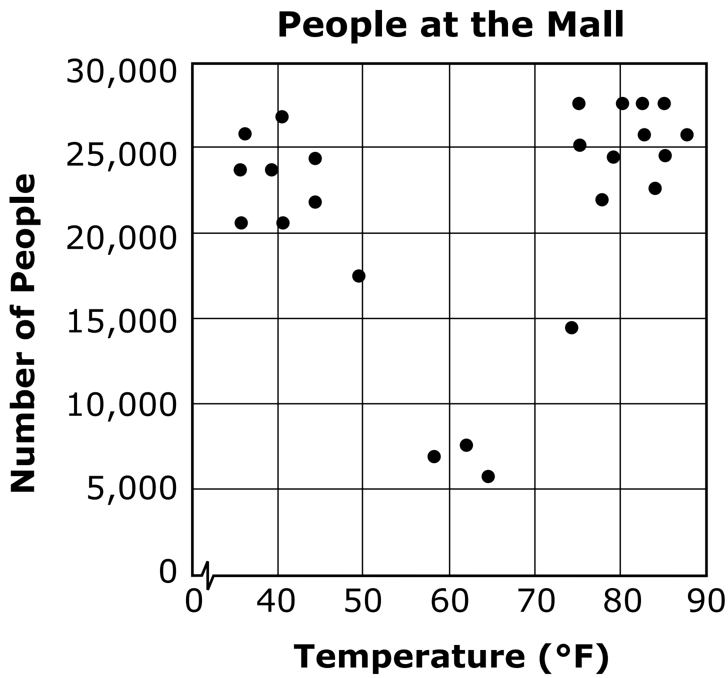
A. **Correct. The student demonstrated an ability to explain how outliers affect measures of central tendency.**

B. The student confused median and range.

C. The student defined median, but this did not answer the question.

D. The student chose an explanation that is not true for this data set, nor did it answer the question.

22 **This scatter plot shows the number of people at a mall each day and the average temperature for the day.**



Based on the scatter plot, which statement is true?

A The number of people at the mall always increases as the temperature rises.

B The number of people at the mall always decreases as the temperature rises.

C Fewer people are at the mall when the temperature is between 70°F and 90°F.

D Fewer people are at the mall when the temperature is between 50°F and 70°F.

Standard: PA.D.1.3 Collect, display, and interpret data using scatter plots. Use the shape of the scatter plot to find the informal line of best fit, make statements about the average rate of change, and make predictions about values not in the original data set. Use appropriate titles, labels and units.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to analyze the scatter plot and then determine which statement about the data is true.

Distractor Rationale:

A. The student does not know how to read the data displayed on the scatterplot.

B. The student saw that the data decreased from the left to the middle.

C. The student does not know how to read the data displayed on the scatterplot or confused fewer and more.

D. Correct. The student demonstrated an ability to interpret data displayed on a scatterplot.

23 **Nancy has a bag of different colored golf balls. Each ball is white, yellow, or green.**

Nancy takes a ball from the bag without looking, records the color, and then returns the ball to the bag. She repeats this process until she has recorded the color of a ball 10 times. Her results are shown.

- **White: 7**
- **Yellow: 2**
- **Green: 1**

Nancy will take another golf ball from the bag without looking. Based on Nancy's data, what is the probability she will take a green golf ball from the bag?

A $\frac{1}{10}$

B $\frac{1}{9}$

C $\frac{1}{7}$

D $\frac{1}{3}$

Standard: PA.D.2.1 Calculate experimental probabilities and represent them as percents, fractions, and decimals between 0 and 1. Use experimental probabilities to predict relative frequencies when actual probabilities are unknown.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the experimental probability of a simple experiment.

Distractor Rationale:

A. Correct. The student demonstrated an ability to calculate experimental probability and represent it as a fraction.

B. The student compared green to non-green.

C. The student compared green to white.

D. The student used 3 for the denominator because there are 3 possible colors.

24

A student randomly selected one marble from a bucket of marbles 50 times with replacement. These are the results of his experiment.

- 5 green marbles
- 8 blue marbles
- 12 red marbles
- 25 purple marbles

Based on these results, what is the probability of selecting a green, blue, red, or purple marble?

Select one button in each row to indicate the expected probability for each color. To make a selection, click a button in one of the four columns, To remove a selected button, click the button again.

Probability of Selecting

Color	50%	0.16	24%	$\frac{1}{10}$
green	○	○	○	○
blue	○	○	○	○
red	○	○	○	○
purple	○	○	○	○

Standard: PA.D.2.1 Calculate experimental probabilities and represent them as percents, fractions, and decimals between 0 and 1. Use experimental probabilities to predict relative frequencies when actual probabilities are unknown.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine the expected probabilities, represented as percents, decimals, and fractions, of selecting different colors in a probability experiment.

Sample Distractor Rationales:

Correct

Probability of Selecting

Color	50%	0.16	24%	$\frac{1}{10}$
green	○	○	○	●
blue	○	●	○	○
red	○	○	●	○
purple	●	○	○	○

Incorrect

Probability of Selecting

Color	50%	0.16	24%	$\frac{1}{10}$
green	●	○	○	○
blue	○	●	○	○
red	○	○	●	○
purple	●	○	○	○

The student focused on the 5 green marbles that were selected and thought this was the same as 50%.

Probability of Selecting

Color	50%	0.16	24%	$\frac{1}{10}$
green	●	○	○	○
blue	○	○	○	●
red	○	●	○	○
purple	○	○	●	○

The student found the number that was close in some way to each result. 5 green means 50% because they both have 5s; 8 blue is closest to the 10 in 1/10; 12 red is close to 16 in 0.16; 25 purple is close to 24%.

25 **School cafeteria workers conduct a survey on Monday to learn how many students will want to buy pizza on Friday.**

What sample should the cafeteria workers choose for the survey?

A every other student who buys lunch

B every teacher who brings a class to lunch

C every fourth student who enters the school

D every other student who brings a lunch from home

Standard: PA.D.2.2 Determine how samples are chosen (randomness) to draw and support conclusions about generalizing a sample to a population, including identifying limitations and biases.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to use a term/definition to identify the best sample for a survey.

Distractor Rationale:

A. The student thought the sample should only include students who buy lunch.

B. The student selected a sample that is not representative of the population.

C. Correct. The student demonstrated an ability to determine the best sample for a survey.

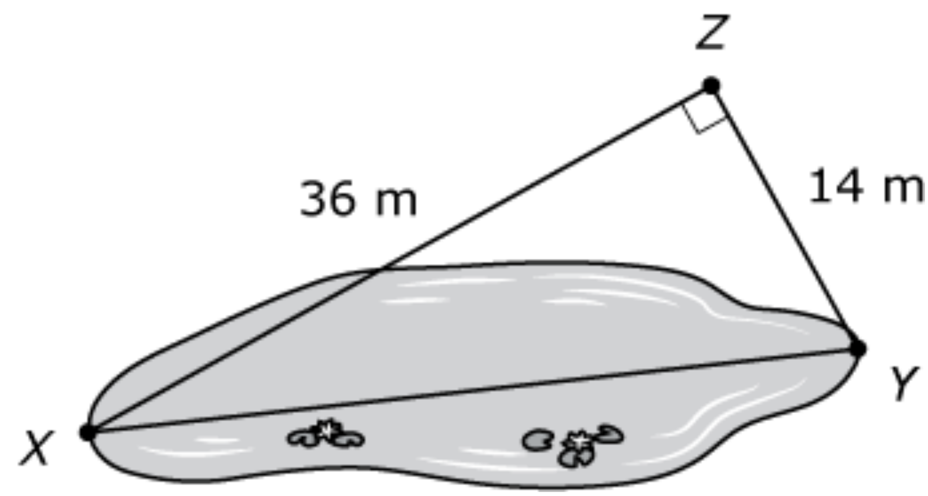
D. The student selects a sample that is not representative of the population.

Cluster Items

The following sample items are part of a cluster. The cluster is presented first and then the two items that follow require use of the cluster. The two items are from different standards.

Use this information to answer the following questions.

Mathew wants to find the length of a pond. He picks three points and records the measurements, as shown in the diagram.



26 **Which measurement is closest to the length of the pond from point *X* to point *Y* in meters?**

A 10 meters

B 22 meters

C 39 meters

D 50 meters

Standard: PA.GM.1.1 Justify the Pythagorean theorem using measurements, diagrams, or dynamic software to solve problems in two dimensions involving right triangles.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use the Pythagorean Theorem to determine the approximate value for the length of the hypotenuse of a triangle.

Distractor Rationale:

A. Balance distractor

B. The student computed 36 − 14.

C. Correct. The student demonstrated an ability to use the Pythaogorean Theorem to solve a problem.

D. The student computed 36 + 14.

27 **Mathew finds the deepest part of the pond to be $\sqrt{185}$ meters. Which measurement describes the depth of the pond?**

A between 13 and 14 meters

B between 14 and 15 meters

C between 92 and 93 meters

D between 93 and 94 meters

Standard: PA.N.1.4 Compare and order real numbers; locate real numbers on a number line. Identify the square roots of perfect squares to 400 or, if it is not a perfect square root, locate it as an irrational number between two consecutive positive integers.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the square root of a number as between two consecutive whole numbers.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to find the square root of a number as between two consecutive positive integers.**
B. Balance distractor
C. The student computed 185 ÷ 2.
D. The student computed 185 ÷ 2 incorrectly.